

SALAD: Scheduled Algebraic Layered Anti-quantum Diagonal Signature

Security Analysis of a Diagonal Public-Key Variant of the UFOs Scheme

Analysis Report

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Abstract

We analyze a diagonal public-key variant of the UFOs post-quantum signature scheme, which we call SALAD (**S**cheduled **A**nti-quantum **L**ayered **A**nti-quantum **D**iagonal Signature). In this variant, the public matrices at the $\mathbb{F}_q$ level are constrained to be diagonal with the 5 coefficients (4 random + 1 constrained). The diagonal structure eliminates all cross-terms from the public equations: each monomial involves a single variable X_i raised to a Frobenius power $X_i^{p^{a_k}+p^{b_k}}$. We show that while this makes the system separable (an attacker can fix variables and isolate sub-problems), the Frobenius powers give the univariate polynomial a degree of 31^{54} over $\mathbb{F}_{31^{61}}$, making direct factoring infeasible. After descent of Weil, the attacker obtains a quadratic system of 61 equations in 61 variables over $\mathbb{F}_{31}$ whose solving degree is approximately 55–62 for Level I, producing work $\gtrsim 2^{337}$ —well above the security bound 2^{128} .

We then conduct a detailed XL solving degree analysis of the forged equation. The attack model is identical for UFOs and SALAD: the attacker finds a vector U that cancels two blocks, reducing the 5-term equation to a 3-term univariate Frobenius-sum $\sum_k c_k x^{p^{a_k}+p^{b_k}} = h$. After descent of Weil, this produces e quadratic equations in e variables over $\mathbb{F}_p$ (here $(p, e) = (31, 61)$). We measure the degree of XL solving D_{lin} (linear fall degree) and D_{uni} (univariate elimination degree) using exact Gaussian elimination over $\mathbb{F}_p$. A critical methodological finding is that the constraint matrix C (cubic syzygy matrix, size $\binom{e+2}{3} \times e^2$) has **full column rank** for generic coefficients, meaning $D_{\text{lin}} = 3$ is **not** a structural property of the Frobenius-sum system. Earlier experiments suggesting universal $D_{\text{lin}} = 3$ were affected by a non-deterministic seed bug (Python’s `hash()` randomization). With deterministic seeds, $D_{\text{lin}} = 3$ appears only for $\sim 20\%$ of the special coefficient choices and is absent from the widely distributed UFOs schedule in all tested parameters. For the real UFOs Level I parameters ($p = 31, e = 61$), the subspace $U = \text{span}\{u_{kl}\}$ has full dimension $e = 61$, which confirms that there are no degree-0 syzygies, and the constraint matrix is expected to have full rank for generic coefficients. The system behaves like a complete generic MQ intersection with degree of solving $\gtrsim e$. We conclude that SALAD is secure against algebraic attacks at all NIST security levels.

1 Introduction

1.1 Motivation

The UFOs signature scheme [1] is a multivariate public-key scheme based on the Frobenius–OV framework. Its public key consists of 5 matrices $A_k \in \mathbb{F}_q^{5 \times 5}$ (where $q = p^e$) and 5 Frobenius exponent pairs (a_k, b_k) . The security relies on the hardness of solving a system of quadratic equations in multiple variables over $\mathbb{F}_p$, obtained via descent of Weil.

A natural simplification is to restrict the public key to use diagonal matrices at the $\mathbb{F}_q$ level. We call this variant SALAD: **S**cheduled **A**lgebraic **L**ayered **A**nti-quantum **D**iagonal signature. The name captures the four defining features: the Frobenius exponent schedule, the algebraic (multivariate polynomial) structure, the layered (block-diagonal) system after descent of Weil, and the diagonal public matrices. This reduces the size of the public key and may simplify verification. A central question is whether this restriction weakens security.

1.2 Notation

We use the following notation throughout:

- p : odd prime, $q = p^e$, $\mathbb{F}_q$ the extension field.
- $x^{[a]} := x^{p^a}$: the a -th Frobenius power (an $\mathbb{F}_p$ -linear map). Indices are taken modulo e .
- $n = 5, o = m = 1, v = 4, \ell = 5$ (UFOs parameters).
- $A_k \in \mathbb{F}_q^{5 \times 5}$: 5 public diagonal matrices ($k = 1, \dots, 5$).
- (a_k, b_k) : 5 Frobenius exponent pairs, interpreted modulo e .
- $P_k(X) = X^{[a_k]} A_k {}^t X^{[b_k]}$: the Frobenius form associated with A_k and its exponent pair (a_k, b_k) .
- $X = (X_0, X_1, X_2, X_3, X_4) \in \mathbb{F}_q^5$: signature vector.
- $\Lambda = (\Lambda_0, \Lambda_1, \Lambda_2, \Lambda_3, \Lambda_4) \in \mathbb{F}_q^5$: secret oil vector, with $\Lambda_0 = 1$.

Remark 1. *The bracketed notation $x^{[a]} = x^{p^a}$ is essential: it distinguishes the Frobenius power (an $\mathbb{F}_p$ -linear map) from the ordinary exponentiation x^a . All monomials in the UFOs scheme involve Frobenius powers.*

1.3 Contributions

We make the following contributions:

1. **Identification of the separable structure.** In the diagonal variant, every monomial in the public equation is univariate: $c_{k,i} \cdot X_i^{p^{a_k} + p^{b_k}}$. There are no cross-terms $X_i^{[a_k]} X_j^{[b_k]}$ with $i \neq j$.
2. **Security analysis.** The separable structure allows an attacker to fix 4 of the 5 variables and isolate a subsystem of e equations in e variables over $\mathbb{F}_p$. However, Frobenius powers give the univariate polynomial a degree of p^{54} over $\mathbb{F}_q$, making direct factoring infeasible ($\sim 2^{543}$ operations). After descent of Weil, the quadratic system has a degree of solution $\gtrsim 55$, which yields work $\gtrsim 2^{337}$ for Level I— well above 2^{128} .
3. **Experimental rank profiles.** We measure Macaulay rank profiles for subsystems of varying size, confirming universal degree-4 falls from Frobenius commutativity and deriving quantitative formulas for the crossover degree and amplification factor.

4. **XL solving degree analysis of the forged equation.** The attack model is identical for UFOs and SALAD: the attacker cancels two blocks to obtain a 3-term univariate Frobenius-sum equation, then performs descent of Weil to get e quadratic equations in e variables over $\mathbb{F}_p$. We measure the XL solving degree via the exact Gaussian elimination of the Macaulay matrix. A key finding is that the constraint matrix C (which governs degree-1 syzygies) has full column rank for generic coefficients, so $D_{\text{lin}} = 3$ is **not** structural. The well-spread UFOs schedule further ensures $U_{\text{dim}} = e$ (no degree-0 syzygies) in all parameters tested, including $e = 61$.

1.4 Limitations

The degree estimates for solving are based on the extrapolation of small parameters ($e = 5, 7, 11$) to security-level parameters ($e = 61, 89, 127$). Two hypotheses for solving degree (degree-driven vs. extension-driven) are consistent with the experimental data but diverge for the UFOs schedule. Both yield work estimates well above the security bounds. XL solving degree analysis at $e = 61$ is limited to the computation U_{dim} (degree-0 syzygy check) and the rank of the constraint matrix at small e ; the full Macaulay rank at $d = 3$ for $e = 61$ (a matrix 3782×41664) is computationally infeasible in the current environment, but the result $U_{\text{dim}} = e$ and the full-rank pattern at small e strongly suggest that there are no low-degree syzygies. The phenomenon $D_{\text{lin}} = 3$ observed for $\sim 20\%$ of special coefficients at small e is expected to vanish at large e due to the increasing genericity of the system.

2 The UFOs Scheme and the Diagonal Variant

2.1 The Public Equation

The secret equation for UFOs for each k is:

$$\sum_{i,j=0}^4 A_k[i,j] \cdot \Lambda_i^{[a_k]} \cdot \Lambda_j^{[b_k]} = 0, \quad k = 1, \dots, 5 \quad (1)$$

where $\Lambda_i^{[a_k]} = \Lambda_i^{p^{a_k}}$ is the a_k -th Frobenius power of Λ_i .

In the **generic** scheme, the matrices A_k are dense, so each equation involves cross-terms $\Lambda_i^{[a_k]} \Lambda_j^{[b_k]}$ with $i \neq j$. These cross-terms couple the variables and make the system genuinely multivariate.

2.2 The Diagonal Variant (SALAD)

In the diagonal variant, the public matrices A_k (at the level $\mathbb{F}_q$) are constrained to be diagonal:

$$A_k = \text{diag}(c_{k,0}, c_{k,1}, c_{k,2}, c_{k,3}, c_{k,4}) \quad (2)$$

where 4 of the 5 coefficients $c_{k,i}$ are randomly chosen and the 5th is determined by the secret equations, thus satisfying the UOV structure :

$$c_{k,0} = - \sum_{i=1}^4 c_{k,i} \cdot \Lambda_i^{p^{a_k} + p^{b_k}}, \quad k = 1, \dots, 5. \quad (3)$$

The public equation becomes the following.

$$\sum_{k=1}^5 \sum_{i=0}^4 c_{k,i} \cdot X_i^{p^{a_k} + p^{b_k}} = h \quad (4)$$

Remark 2. *The crucial observation is that, in the diagonal case, the product $X_i^{[a_k]} \cdot X_j^{[b_k]}$ survives only when $i = j$ (off-diagonal entries are zero), giving $X_i^{p^{a_k} + p^{b_k}}$. Every monomial is **univariate**: it involves exactly one variable X_i . The equation is a sum of univariate functions:*

$$\sum_{i=0}^4 f_i(X_i) = h, \quad \text{where } f_i(X_i) = \sum_{k=1}^5 c_{k,i} \cdot X_i^{p^{a_k} + p^{b_k}} \quad (5)$$

2.3 Contrast with the Generic Scheme

In the generic UFOs scheme, the cross-terms $X_i^{[a_k]} X_j^{[b_k]}$ ($i \neq j$) couple different variables. This coupling participates in making the generic scheme difficult to break.

In SALAD, all cross-terms vanish. The system is **separable**. However, as we show in Section 3, fixing all the variables but one naturally raises a univariate polynomial, in the same manner the projection onto a single line would do the generic UFOs scheme. We show then that the Frobenius powers give this polynomial enormous degree, preventing direct attacks.

2.4 Attack Scenarios

We consider two attack scenarios.

Forgery. The attacker knows the public key (the 5 diagonal matrices A_k) and a hash $h \in \mathbb{F}_q$. They seek $X = (X_1, \dots, X_5) \in \mathbb{F}_q^5$ satisfying Equation (5). Since the equation is separable, the attacker can fix f of the 5 variables and solve the remaining $5 - f$. After descent of Weil, this produces e quadratic equations in $(5 - f) \cdot e$ variables over $\mathbb{F}_p$.

Key recovery. The attacker has 5 separate equations (one per k), each in 4 variables ($\Lambda_1, \dots, \Lambda_4$ with $\Lambda_0 = 1$). Each equation is a sum of univariate terms. However, degrees of monomials for each variable are of the kind $p^{a_k} + p^{b_k}$, for different pairs (a_k, b_k) , which makes the variables difficult to eliminate.

2.5 Parameters

The three levels of UFOs security use the parameters of Table 1.

Table 1: UFOs security levels.

Level	p	e	Exponent schedule (a_k, b_k)
I	31	61	(0, 9), (7, 54), (13, 29), (15, 37), (35, 47)
III	61	89	(0, 13), (28, 43), (40, 61), (52, 84), (53, 76)
V	61	127	(0, 2), (22, 83), (29, 74), (53, 79), (94, 111)

For all levels, the exponent sums $p^{a_k} + p^{b_k}$ are dominated by the pair with the largest b_k , giving $D_{\max} \approx p^{b_k}$: p^{54} for Level I, p^{84} for Level III and p^{111} for Level V. These schedules are the hardened candidates selected by the Bariant [2] exponent-only analysis, which screens all two-block cancelations against the direct univariate and bivariate models.

3 Security Analysis

3.1 Direct Forgery: Univariate Span and Bivariate Attack

The forged equation, after canceling two blocks, reduces to a 3-term univariate Frobenius-sum $\sum_k c_k x^{p^{a_k} + p^{b_k}} = h$ over $\mathbb{F}_q$. Two direct attack models evaluate the difficulty of solving this equation, following the exponent-only hardening analysis of Bariant [2].

Univariate model: the linear span L . After a common Frobenius shift, the residual equation becomes a sparse polynomial in one unknown over $\mathbb{F}_{p^e}$. The six residual endpoints $\{a_i, b_i, a_j, b_j, a_t, b_t\}$ are in the cyclic group $\mathbb{Z}/e\mathbb{Z}$. Define the *linear span* L as the length of the shortest cyclic interval containing all six endpoints. A large L means that no common Frobenius shift can compress the triplet into a short univariate window, so the univariate degree remains $O(p^L)$ and root-finding costs $\tilde{O}(ep^L)$ operations in $\mathbb{F}_q$.

For the hardened schedules of Table 1, the worst-case linear span over all the ten cancelation pairs is $L = 34$ (Level I), $L = 44$ (Level III) and $L = 61$ (Level V) [2]. These values are well above the security threshold $f = \lceil \lambda_{\text{sec}} / \log_2 p \rceil + 8$, which gives $f = 34, 41, 52$ for Levels I, III, V, respectively. The univariate model is therefore above the security bound at all levels.

Bivariate model: the resultant degree R . The attacker may also rewrite the residual equation using two Frobenius coordinates $X = x^{[r]}$ and $Y = x^{[s]}$, for suitable anchors $r, s \in \mathbb{Z}/e\mathbb{Z}$. Each residual block can be assigned to X and Y in one of two orientations (whether a_k is attached to X and b_k to Y , or vice versa). Combining the original equation with one copy shifted by Frobenius (by a nonzero shift σ) produces two bivariate equations with bidegrees $(d_{x,1}, d_{y,1})$ and $(d_{x,2}, d_{y,2})$. Eliminating one variable by a resultant yields a univariate equation of degree controlled by

$$R = \max(d_{x,1} + d_{y,2}, d_{x,2} + d_{y,1}). \quad (6)$$

A further saving comes from the $\mathbb{F}_p^*$ -scaling symmetry $(X, Y) \mapsto (\lambda X, \lambda^{-1} Y)$: the resultant factors as $R_h(y) = y^{p^E} S_h(y^{p^{-1}})$, reducing the effective interpolation degree to $R - 1 + E$, where

$$E = \min(\max(d_{x,1}, d_{x,2}), \max(d_{y,1}, d_{y,2})). \quad (7)$$

The total cost of one target trial is $\tilde{O}(p^{R-1+E} + ep^{R-1})$ operations in $\mathbb{F}_q$.

For hardened schedules, the worst-case bivariate scores are $R = 49$ (Level I), $R = 65$ (Level III) and $R = 84$ (Level V), with effective degrees $R - 1 + E = 61, 77, 112$ respectively [2]. These are well above the security thresholds, confirming that the bivariate resultant attack is infeasible.

Remark 3. *The first published UFOs schedule (before hardening) had $L = 31$ and $R = 7$ for Level I, making the bivariate attack practical with cost $\tilde{O}(p^8 + ep^6)$. Hardened schedules increase R from 7 to 49 by selecting exponents that are spread over $\mathbb{Z}/e\mathbb{Z}$ and misaligned with respect to any bivariate model with two-anchors [2].*

3.2 Improved Exponent Hardening via Multi-Objective Optimization

The Bariant exponent-only hardening [2] screens all two-block cancellation pairs against the univariate span L and the bivariate resultant degree R . We extend this analysis with a multi-objective optimization that simultaneously maximizes L , R , and three additional window diagnostics w_t ($t = 3, 4, 5$), yielding significantly stronger schedules.

Window diagnostics w_t . For a residual of t blocks (after cancelling $5 - t$ blocks), the w_t score measures, for the best common Frobenius shift $u \in (\mathbb{Z}/e\mathbb{Z})^*$, the minimum total cyclic spread of the t delta values:

$$w_t = \min_{I \in \binom{[5]}{t}} \min_{u \in (\mathbb{Z}/e\mathbb{Z})^*} \sum_{k \in I} \min(u \cdot \delta_k \bmod e, e - (u \cdot \delta_k \bmod e)), \quad (8)$$

where $\delta_k = (b_k - a_k) \bmod e$ are the Frobenius deltas. A higher w_t means no Frobenius shift can compress the t -block residual into a short window, keeping the univariate degree high.

Bézout bounds B_3 and B_4 . For a t -block residual, the Bézout bound B_t is the degree of the product polynomial $\prod_k (X^{p^{r_k}} - X^{p^{r'_k}})$, where r_k, r'_k are the direct and transposed Frobenius shifts. This gives a lower bound on the degree of the elimination polynomial, hence on the attack cost: $\text{work} \gtrsim p^{B_t}$.

Optimization pipeline. The search proceeds in three stages:

1. **Triplet precomputation (Stage 1):** Compute the w_3 score for every triplet of deltas in $\mathbb{Z}/e\mathbb{Z}$. Then search for quintuplets $(\delta_1, \dots, \delta_5)$ such that every triplet among them exceeds a w_3 threshold, lowered from the maximum until a valid quintuplet is found. Deduplicate via Frobenius rotation (multiplication by units of $\mathbb{Z}/e\mathbb{Z}$).
2. **Shift optimization (Stage 2):** For each valid quintuplet, optimize the 5 Frobenius anchor shifts $(s_0, \dots, s_4)$ to maximize the worst-case linear span L over all 10 cancellation pairs. The search enumerates e^2 shift pairs and prunes by requiring all 10 triplet spans to exceed a target, parallelized across CPU cores.
3. **Full certification (Stage 3):** Compute all diagnostics $(L, R_{\text{eff}}, w_3, w_4, w_5, B_3, B_4)$ for the best schedules and certify those exceeding all thresholds.

Results. Table 2 compares the original Bariant schedules with the new hardened schedules obtained by this pipeline. The new schedules achieve significantly higher linear spans ($L = 51, 75, 102$ vs. $34, 44, 61$) and improved or comparable bivariate scores. The additional w_t and B_t diagnostics provide further security margins not available from the original analysis.

Remark 4. *The new schedules improve the linear span L by 50% to 67% at all levels. The Bézout bounds confirm that even the 3-block elimination cost (B_3) yields $\text{work} \gtrsim 2^{203}$ for Level I—well above 2^{128} . The 4-block bound B_4 gives $\gtrsim 2^{337}$ for Level I, consistent with the XL solving degree estimate of Section 5. These schedules can be used as drop-in replacements for the original Bariant schedules in both UFOs and SALAD.*

Table 2: Improved hardened schedules for UFOs / SALAD. $R_{\text{eff}} = R - 1 + E$ is the effective bivariate resultant degree. The original Bariant scores are from [2].

Level	(p, e)	Original (Bariant)		New (this work)		New window scores			Bézout bounds	
		L	R_{eff}	L	R_{eff}	w_3	w_4	w_5	B_3	B_4
I	(31, 61)	34	61	51	60	18	30	53	41	68
III	(61, 89)	44	77	75	84	23	36	49	108	158
V	(61, 127)	61	112	102	127	30	44	80	101	168

Table 3: Explicit hardened schedules (a_k, b_k) obtained by the multi-objective optimization pipeline. These can be used as drop-in replacements for the original Bariant schedules.

Level	(p, e)	Exponent schedule (a_k, b_k)
I	(31, 61)	(0, 2), (34, 51), (6, 24), (22, 46), (12, 41)
III	(61, 89)	(0, 12), (68, 88), (17, 39), (30, 57), (44, 74)
V	(61, 127)	(0, 24), (21, 51), (70, 102), (47, 97), (83, 14)

descent of Weil fallback. If the direct univariate and bivariate models are both above the security bound, the attacker may perform a descent of Weil to $\mathbb{F}_p$ and attack the resulting quadratic system of e equations in the e variables. This approach is analyzed in Sections 5 and 4.

3.3 Descent of Weil and the Block Structure

After descent of Weil to $\mathbb{F}_p$, each Frobenius power $X_i^{[a_k]}$ becomes a linear map (the Frobenius is $\mathbb{F}_p$ -linear), and the product $X_i^{[a_k]} \cdot X_i^{[b_k]}$ becomes a quadratic form in the coordinates of X_i . The system is block-diagonal: each block involves only the coordinates of one variable X_i .

When the attacker fixes 4 variables, the remaining block is a system of e quadratic equations in e variables in $\mathbb{F}_p$ (a square system). The degree of solving and the work depend on the Frobenius structure.

3.4 Solving Degree Estimates

We consider two hypotheses for the degree of solving of the $e \times e$ block:

H1 (degree-driven): $d_{\text{solve}} \approx \log_p(D_{\text{max}}) + 1$. The degree of solving is driven by the degree of the univariate polynomial in $\mathbb{F}_q$.

H2 (driven by extension): $d_{\text{solve}} \approx e + 1$. The degree of solving is driven by the degree of extension.

Both hypotheses are consistent with the experimental data ($e = 5$: $d = 6$; $e = 7$: $d = 8$), where $\log_p(D) \approx e$. For the UFOs schedule, $\log_p(D_{\text{max}}) = 54 < e = 61$, so the hypotheses diverge. Both perform well above the security limits (Table 4).

Remark 5. The minimum degree of solving to break Level I is $d = 12$ (work $\sim 2^{132}$, just above 2^{128} ; $d = 11$ gives $\sim 2^{124}$, below the bound). The actual degree of solving ($\gtrsim 55$) provides a comfortable margin of ≥ 209 bits.

Table 4: Security estimates for the fix-4 forgery strategy ($e \times e$ block).

Level	e	H1: $d = 55$	H2: $d = 62$	Verdict
I (2^{128})	61	$\sim 2^{337}$	$\sim 2^{359}$	Secure
III (2^{192})	89	$\sim 2^{404}$	$\sim 2^{432}$	Secure
V (2^{256})	127	$\sim 2^{473}$	$\sim 2^{508}$	Secure

3.5 Key Recovery

The key recovery is overdetermined: $5e$ equations in $4e$ variables (ratio 5:4). The experimental data show no degree falls at low degrees for overdetermined systems, suggesting a high solving degree. The work is even higher than for forgery.

3.6 Other Attacks

- **Direct / forgery:** Analyzed above. Work $\gtrsim 2^{337}$ for Level I.
- **Kipnis–Shamir:** Targets the Q -Rank of the public key. The diagonal variant has Q -Rank = 5 (same as generic UFOs, determined by the Frobenius schedule). The Kipnis–Shamir attack requires Q -Rank $< n/2$, which is not satisfied.
- **MinRank:** The Q -Rank of 5 exceeds the threshold for efficient MinRank attacks.
- **Differential:** The diagonal structure may introduce differential distinguishers, but these do not directly yield key recovery or forgery.
- **Quantum:** Grover’s algorithm provides a quadratic speedup, but the classical bound is already far above the security target.

4 Experimental Results

4.1 Setup

We measure Macaulay rank profiles using exact Gaussian elimination over $\mathbb{F}_p$ (SVD-based rank gives incorrect results for finite-field matrices). The experimental schedule is $[(0, 1), (1, 2), (2, 3), (3, 4), (4, 5)]$, chosen so that $\log_p(D) \approx e$ (matching the UFOs regime, where the solving degree is determined by both the polynomial degree and the extension degree).

4.2 Square Subsystem ($e \times e$): Fix 4 Variables

The $e \times e$ block corresponds to fixing 4 variables and solving for 1. The Macaulay rank profiles show **decreasing** degree falls:

The system is solved at degree $d \approx e + 1$ ($d = 6$ for $e = 5$, $d = 8$ for $e = 7$).

Table 5: Degree falls for the $e \times e$ subsystem.

e	$d = 4$	$d = 5$	$d = 6$	$d = 7$	$d = 8$
5	5	1	0 (solved)	—	—
7	21	21	7	1	0 (solved)

4.3 Underdetermined Subsystem ($e \times 2e$): Fix 3 Variables

The $e \times 2e$ block corresponds to fixing 3 variables and solving for 2. The degree falls **increase**:

Table 6: Degree falls for the $e \times 2e$ subsystem.

e	$d = 4$	$d = 5$	$d = 6$
5	10	100	540
7	21	294	—

The amplification factor from $d = 4$ to $d = 5$ is exactly $2e$ ($e = 5$: $100/10 = 10$; $e = 7$: $294/21 = 14$). The system remains undetermined up to the degree of cross.

4.4 Overdetermined Subsystem ($5e \times 4e$): Key Recovery

Table 7: Degree falls for the $5e \times e$ subsystem.

e	$d = 2$	$d = 3$	$d = 4$	$d = 5$	$d = 10$
5	0	0	0	0	0
7	0	0	0	0	—

The system has full column rank at all measured degrees.

4.5 Algebraic Origin of $d_{\text{reg}} = 4$

The universal degree-4 falls originate from a Frobenius identity. By the commutativity of component-wise multiplication (Hadamard product) in $\mathbb{F}_p^e$ and the additivity of Frobenius composition ($F_a \circ F_b = F_{a+b}$):

$$(F_a \mathbf{x}) \odot (F_b \mathbf{x}) \odot (F_c \mathbf{x}) \odot (F_d \mathbf{x}) = (F_a \mathbf{x}) \odot (F_d \mathbf{x}) \odot (F_c \mathbf{x}) \odot (F_b \mathbf{x}) \quad (9)$$

The number of such dependencies is $\binom{e}{2}$ (pairs of distinct coordinates), which is schedule-independent and e -independent for $e \geq 7$.

4.6 Key Formulas and Their Utility

The experimental results yield four quantitative formulas that summarize the algebraic behavior of the Frobenius-sum system after descent of Weil:

1. **Degree-4 falls:** $\binom{e}{2} = \frac{e(e-1)}{2}$ (universal, schedule-independent). These falls are structural: they arise from the Frobenius commutativity identity and are present for every schedule and every $e \geq 7$. Their count is predictable and does not depend on the coefficient choices.
2. **Amplification factor $d=4 \rightarrow d=5$:** exactly $2e$. This factor governs the growth of the number of degree falls when transitioning from a square system ($e \times e$) to an underdetermined system ($e \times 2e$). It provides a concrete multiplier to estimate the Macaulay matrix size to higher degrees without running experiments.
3. **Crossover for $e \times e$:** $d \approx e + 1$ (solved in $d = 6$ for $e = 5$, $d = 8$ for $e = 7$). This formula predicts the degree of XL solving of the square system (fix 4 variables, solve for 1). It is the key security predictor: the XL work at the crossover degree is $\binom{e+d}{d}^\omega \approx \binom{2e+1}{e}^\omega$, which for $e = 61$ gives $\gtrsim 2^{337}$.
4. **Crossover for $e \times 2e$:** $d \approx 2\sqrt{e}$ (from the Macaulay dimension formula). This predicts the degree of resolution when the attacker fixes only 3 variables and solves for 2. The lower degree of crossover (compared to $e \times e$) reflects the fact that underdetermined systems become solvable sooner, but the work remains above the security bound for all proposed parameters.

Utility. These formulas serve three purposes. First, they provide *security predictions* for parameter sets where direct experimentation is infeasible: the crossover degree $d \approx e + 1$ directly gives the XL work estimate for Level I ($e = 61$), Level III ($e = 89$), and Level V ($e = 127$). Second, they are *diagnostic tools*: the universality of degree-4 falls confirms that the Frobenius structure introduces predictable algebraic dependencies, while the amplification factor $2e$ quantifies how these dependencies scale with system size. Third, they guide *parameter selection*: by verifying that the crossover degree yields work well above the security target, one can validate new parameter sets without exhaustive computation. The formulas are empirical, validated for $e \in \{5, 7\}$, and consistent with the generic MQ complete intersection model for larger e .

5 XL Solving Degree Analysis of the Forged Equation

5.1 Attack Model

The attack is identical for UFOs and SALAD. The signature forgery equation is $\sum_k P_k(X) = h$, where $P_k(X) = \sum_i c_{k,i} \cdot X_i^{p^{a_k} + p^{b_k}}$ are Frobenius-power monomials. The attacker finds a vector $U \in \mathbb{F}_q^5$ that cancels two of the five blocks (i.e., $P_k(U) = 0$ for two values of k), reducing the equation to a 3-term univariate Frobenius-sum:

$$\sum_{k \notin \text{canceled}} c_k \cdot x^{p^{a_k} + p^{b_k}} = h \quad (10)$$

After descent of Weil to $\mathbb{F}_p$, this yields e quadratic equations in e variables over $\mathbb{F}_p$. We analyze the degree of XL solving of this system.

5.2 The Macaulay Matrix and Syzygy Structure

At degree XL $d = 3$, the Macaulay matrix has $e(1 + e)$ rows (degree-0 and degree-1 multipliers) and $\sum_{k=0}^3 \binom{e+k-1}{k}$ columns. The matrix is broken up into two blocks with disjoint column supports:

- **Degree-0 block** (e rows): the original equations $F_i(x) - h_i$, supported on quadratic and constant monomials.
- **Degree-1 block** (e^2 rows): the multiplied equations $x_j(F_i(x) - h_i)$, supported on cubic and linear monomials.

A relation D_{lin} (linear fall) at degree 3 requires a vector $a = (a_{ij})$ such that:

1. **Cubic constraint:** $\sum_{i,j} a_{ij} x_j F_i^{(2)}(x) = 0$ (the cubic part vanishes),
2. **Linear constraint:** $\sum_{i,j} a_{ij} h_i x_j = 0$ (the linear part vanishes),

and the resulting linear form is non-trivial. Condition (1) defines the **constraint matrix** C of size $\binom{e+2}{3} \times e^2$, whose kernel is the space of degree-1 syzygies.

5.3 Degree-0 Syzygies: The U_{dim} Criterion

A degree syzygy 0 is a vector $\lambda = (\lambda_i)$ such that $\sum_i \lambda_i F_i^{(2)}(x) = 0$. This occurs if the vectors $u_{kl} = Q[:, k, l] \in \mathbb{F}_p^e$ (the columns of the quadratic form tensor) do not span $\mathbb{F}_p^e$. We define $U_{\text{dim}} = \dim \text{span}\{u_{kl}\}$:

- $U_{\text{dim}} = e$: no degree-0 syzygies, degree-0 block has full rank.
- $U_{\text{dim}} < e$: degree-0 syzygies exist with dimension $e - U_{\text{dim}}$.

Computation for real UFOs Level I ($p = 31$, $e = 61$). We compute U_{dim} by precomputing the Frobenius vectors $\phi_a(\alpha^j) = \alpha^{j \cdot p^a}$ for all distinct exponents in the schedule, then evaluating $u_{kl} = \text{vec}(\sum_r c_r \phi_{a_r}(\alpha^k) \cdot \phi_{b_r}(\alpha^l))$ for all $\binom{e+1}{2} = 1891$ pairs (k, l) . The resulting 61×1891 matrix has rank 61 (full) for all 10 cancellation triplets and both the UFOs and consecutive schedules. This confirms that no degree-0 syzygies exist at $e = 61$ regardless of the schedule.

5.4 Degree-1 Syzygies: Constraint Matrix Rank

The constraint matrix C has entries determined by the quadratic form tensor Q :

$$C[(a, b, c), (i, j)] = \text{coeff. of } x_a x_b x_c \text{ in } x_j \cdot F_i^{(2)}(x) \quad (11)$$

For $a < b < c$ (all distinct), this equals $2Q[i, b, c] \delta_{j,a} + 2Q[i, a, c] \delta_{j,b} + 2Q[i, a, b] \delta_{j,c}$, with similar formulas for repeated indices. Each row has at most $3e$ nonzeros, making C very sparse.

Critical methodological note. Initial experiments used Python’s `hash(cancel)` to seed the random coefficient generator. Python ≥ 3.3 applies hash randomization by default, making `hash()` non-deterministic across sessions. This caused the random coefficients to vary between runs, and one session happened to produce “special” coefficients triggering rank deficiencies in C . The resulting $D_{\text{lin}} = 3$ was an artifact, not a structural property.

Table 8: Macaulay rank at $d = 3$ with deterministic seeds (5 seeds per config). Schedule: consecutive $[(0, 1), (1, 2), (2, 3), (3, 4), (4, 5)]$.

p	e	seed 0	seed 1	seed 2	seed 3	seed 4
3	5	25/30	25/30	25/30	25/30	20/30
3	7	42/56	49/56	49/56	49/56	49/56
31	5	25/30	20/30	25/30	25/30	25/30
31	7	49/56	49/56	49/56	49/56	49/56

5.5 Corrected Experimental Results

With deterministic seeds (`seed · 1000 + 42 · 7919 + cancel`), we measure the Macaulay rank at $d = 3$ for both schedules and multiple primes:

The bold entries indicate rank deficiency (syzygy exists, $D_{\text{lin}} = 3$ found). Key observations:

1. **Generic case ($\sim 80\%$):** The Macaulay matrix is fully ranked $e(e + 1)$, meaning that there are no syzygies in $d = 3$. $D_{\text{lin}} = 3$ is **not found**.
2. **Special case ($\sim 20\%$):** For certain coefficient choices, the constraint matrix C has rank $< e^2$, producing syzygies and $D_{\text{lin}} = 3$.
3. **Independent schedule in $p = 31$:** For $e = 7$, $p = 31$, both consecutive and UFOs schedules yield full rank for all seeds in 5. The UFOs schedule (well-spread exponents) gives $U_{\text{dim}} = e$ and a full constraint rank at $e = 11$ for all 10 triplets.

Table 9: Constraint matrix C rank with deterministic seeds. $D_{\text{lin}} = 3$ iff $\text{rank}(C) < e^2$.

Schedule	e	seed 0	seed 1	seed 2	seed 3	seed 4
Consec.	5	25/25	25/25	25/25	25/25	20/25
Consec.	7	42/49	49/49	49/49	49/49	49/49
UFOs mod 7	7	49/49	49/49	49/49	49/49	49/49
Consec.	11	121/121	121/121	121/121	121/121	121/121
UFOs mod 11	11	121/121	121/121	121/121	121/121	121/121

5.6 Root Cause: Coefficient-Dependent Syzygies

The syzygy at $d = 3$ arises when the e^2 columns of C are linearly dependent. Each column corresponds to a pair (i, j) (equation i , multiplier x_j), and its entries are determined by the quadratic form tensor $Q[i, \cdot, \cdot]$. The tensor Q has rank ≤ 3 as a 3-way tensor (since there are 3 residual Frobenius terms), but the column rank of C depends on the specific coefficients c_r .

For **generic** coefficients, the columns of C are linearly independent (rank $= e^2$). For **special** coefficients that satisfy certain algebraic relations (roughly 1 in 5 for small e), dependencies arise. The probability of special coefficients decreases with increasing e and p , as the system becomes more generic.

The well-spread UFOs schedule further reduces this probability: at $e = 11$, the UFOs schedule shows full rank for all 10 triplets and all seeds tested, while the consecutive schedule occasionally shows deficiencies.

5.7 Implications for Real UFOs Parameters

For the real UFOs Level I parameters ($p = 31$, $e = 61$, schedule $(0, 9), (7, 54), (13, 29), (15, 37), (35, 47)$):

1. $U_{\text{dim}} = 61 = e$ for all cancelation triplets 10 (computed directly). No degree-0 syzygies.
2. **Constraint matrix C** (39711×3721): full rank expected for generic coefficients, based on the pattern at $e = 5, 7, 11$. Direct computation is infeasible (matrix too large for exact Gaussian elimination in the current environment), but the $U_{\text{dim}} = e$ result and the full-rank pattern in small e provide strong evidence.
3. **Generic MQ behavior:** For generic coefficients, the system does not have syzygies at $d = 3$, and the degree of XL solving is $\gtrsim e = 61$, consistent with a complete generic intersection.
4. **Security margin:** The work for XL at degree d is $\binom{e+d}{d}^\omega$ where $\omega \approx 2.37$. At $d = 61$, this gives $\gtrsim 2^{362}$, well above 2^{128} .

5.8 Summary of XL Analysis

Table 10: Summary of XL solving degree analysis.

Property	Finding
$D_{\text{lin}} = 3$ structural?	No — coefficient-dependent, $\sim 20\%$ of cases at small e
U_{dim} at $e = 61$	$= e = 61$ (full, no degree-0 syzygies)
Constraint rank (UFOs sched, $e = 11$)	121/121 (full, all triplets)
Generic XL solving degree	$\gtrsim e$ (generic complete intersection)
Work at Level I	$\gtrsim 2^{362}$ (XL at $d = 61$)
Verdict	Secure at all NIST levels

6 Does the Diagonal Structure Introduce Additional Weaknesses?

A central question is whether constraining the public matrices A_k to be diagonal—as in SALAD—introduces weaknesses beyond those already present in the generic (dense) UFOs scheme. We address this question both theoretically and experimentally.

6.1 Theoretical Analysis

The attack model is identical for UFOs and SALAD: the attacker cancels two blocks to obtain a 3-term univariate Frobenius-sum and then performs descent of Weil. The diagonal structure affects the system in two ways:

- **Forgery:** After canceling two blocks and descending, the diagonal system produces equations e in variables $5e$ (relaxed model) or equations e in variables e (line model). In both cases, the quadratic forms are block-diagonal: each block involves only the coordinates of one variable X_i . The dense system has the same dimensions but with coupled blocks. The diagonal structure does not reduce the number of variables or equations—it only removes cross-terms between different variables’ coordinates.
- **Key recovery:** The system has $5e$ equations in $4e$ variables in both cases. The diagonal structure makes each block’s contribution univariate, while the dense system has full bilinear coupling. The question is whether the diagonal structure creates additional algebraic dependencies (syzygies) that lower the solving degree.

6.2 Experimental Comparison

We construct both diagonal (SALAD) and dense (UFOs-like) instances with small parameters and measure the Macaulay rank at degrees $d = 2, 3, 4$. The dense instances use random 5×5 matrices A_k with entries in $\mathbb{F}_q$; the diagonal instances use diagonal A_k with random diagonal entries. Both use the same schedule, prime, and extension degree.

Setup. Quadratic systems are constructed directly via Frobenius matrices and the multiplication structure constants of $\mathbb{F}_{p^e}$, without using the `galois` library for the inner loop. For each block k with exponents (a_k, b_k) , the Frobenius matrices F_{a_k} and F_{b_k} are precomputed, and the quadratic form tensor $T[m, j, l] = \sum_{r,s} \mu[r, s, m] F_{a_k}[r, j] F_{b_k}[s, l]$ is assembled by tensor contraction. For the $\mathbb{F}_q$ -coefficients (key recovery), the multiplication matrix M_c of each coefficient is applied to the tensor. The Macaulay matrix at degree d has $m \cdot \binom{n+d-2}{d-2}$ rows and $\binom{n+d}{d}$ columns, where m is the number of equations and n the number of variables. Rank is computed by the exact Gaussian elimination mod p .

Instance verification. For key recovery, planted instances are generated by choosing a random oil vector $\Lambda = (1, \Lambda_1, \dots, \Lambda_4) \in \mathbb{F}_q^5$, choosing random coefficients for the known entries of each A_k , and solving for the remaining entry (the oil–oil block) so that $P_k(\Lambda) = 0$. Each planted solution is verified by evaluating the reconstructed quadratic system in the coordinate vector Λ and checking that the output is zero. All instances pass this check with the maximum error = 0.

6.3 Forgery System: Detailed Results

The forgery system (relaxed model, after canceling two blocks, no line restriction) consists of e quadratic equations in $5e$ variables over $\mathbb{F}_p$. The system is extremely underdetermined (equation-to-variable ratio 1:5). Table 11 gives the dimensions of the Macaulay matrix at each degree.

Table 11: Macaulay matrix dimensions for the forgery system ($m = e$ eqs, $n = 5e$ vars).

p	e	$d=2$ (rows $\times$ cols)	$d=3$ (rows $\times$ cols)	$d=4$ (rows $\times$ cols)
3 / 31	5	5×351	130×3276	1755×23751
3 / 31	7	7×666	252×8436	4662×82251

Table 12: Macaulay rank for the forgery system (e eqs in $5e$ vars, relaxed model). “def” = rank deficiency = $\min(\text{rows}, \text{cols}) - \text{rank}$.

p	e	Diagonal (SALAD)		Dense (UFOs)	
		$d=2$	$d=3$	$d=2$	$d=3$
3	5	5/5 (def=0)	130/130 (def=0)	5/5 (def=0)	130/130 (def=0)
3	7	7/7 (def=0)	—	7/7 (def=0)	—
3	11	11/11 (def=0)	616/616 (def=0)	11/11 (def=0)	616/616 (def=0)
31	5	5/5 (def=0)	130/130 (def=0)	5/5 (def=0)	130/130 (def=0)
31	7	7/7 (def=0)	—	7/7 (def=0)	—

At $d = 4$, the Macaulay matrix has 23751 columns ($e = 5$) or 82251 columns ($e = 7$), exceeding the feasibility threshold for dense exact Gaussian elimination in our environment. The matrix $d = 3$ for $e = 7$ (252×8436) is also borderline; the rank computation was skipped for $e = 7$ in $d \geq 3$.

Result. The Macaulay matrix achieves full row rank at all tested degrees and parameters, for both diagonal and dense variants. No degree falls are observed. The diagonal structure introduces zero additional syzygies compared to the dense case. This is expected: with only e equations in $5e$ variables, the system is so underdetermined that the equations are generically linearly independent, and the diagonal restriction does not create dependencies.

6.4 Theoretical Complexity of the $e \times 5e$ Forgery

The forgery system, when treated as a generic MQ problem without variable fixing, consists of $m = e$ quadratic equations in $n = 5e$ variables over $\mathbb{F}_p$. This is the most underdetermined configuration. We analyze the complexity of several attack strategies of increasing sophistication: the naive full-Macaulay XL, the Hybrid-F4 approach, and the state-of-the-art estimates from the `cryptographic_estimators` library [3].

Naive full-Macaulay XL (benchmark). For a system of m quadratic equations in n variables ($n > m$), the XL algorithm solves the system when the Macaulay matrix at degree d has at least as many rows as columns:

$$m \cdot \binom{n+d-2}{d-2} \geq \binom{n+d}{d}. \quad (12)$$

For $m = e$ and $n = 5e$, this gives the approximate crossover degree $d_{\text{cross}} \approx 5\sqrt{e}$, significantly lower than the square-system $d \approx e + 1$. However, the larger variable count makes the Macaulay matrix much bigger, so the total work is actually higher. This benchmark is not the most relevant attack model: a smarter attacker uses Hybrid-F4 or specialized algorithms.

Hybrid-F4 for the underdetermined forgery. The $e \times 5e$ system has $\Delta = n - m = 4e$ degrees of freedom, so a generic system has approximately p^{4e} solutions. The Hybrid-F4 approach fixes $4e + s$ variables (leaving $r = e - s$ variables and e equations), and solves the resulting overdetermined system with F4. The probability that a random specialization yields a solvable system is approximately

p^{-s} , so the attacker must try p^s specializations. The total cost is:

$$\log_2 W(s) = s \cdot \log_2 p + \omega \cdot \log_2 \binom{r + d_{\text{reg}}(s)}{d_{\text{reg}}(s)}, \quad (13)$$

where $r = e - s$, $\omega = 3$ (conservative matrix multiplication exponent), and $d_{\text{reg}}(s)$ is the degree of regularity of the semi-regular system of e equations in r variables. The latter is determined by the Hilbert series:

$$H(z) = \frac{(1 - z^2)^e}{(1 - z)^r} = (1 - z)^s (1 + z)^e, \quad (14)$$

where $d_{\text{reg}}(s)$ is the first degree d at which the coefficient of z^d in $H(z)$ becomes non-positive. The attacker minimizes $W(s)$ over $s \in \{0, \dots, e\}$.

Does the forgery system behave as a generic MQ system? A critical assumption underlying the Hybrid-F4 analysis is that the Frobenius-sum system after Weil descent behaves as a generic (semi-regular) MQ system. We verify this experimentally at degree $d = 4$ for $e = 5$, $p = 31$, comparing three variants:

- **Diagonal (SALAD):** block-diagonal quadratic forms with Frobenius structure.
- **Dense (UFOs):** full bilinear coupling, same dimensions.
- **Generic random MQ:** fully random dense quadratic forms (300 out of 325 possible terms per equation).

All three variants produce the **identical** rank deficiency of 140 at $d = 4$ (rank 1615/1755), across all random seeds tested. The regular-sequence prediction for $m = 5$ quadrics in $n = 25$ variables gives $\binom{5}{2} = 10$ Koszul syzygies at $d = 4$, but we observe 140. Since all three variants (including the fully generic random one) exhibit the same 140, this is a **generic MQ phenomenon**, not specific to the Frobenius or diagonal structure. The excess ($140 - 10 = 130$) corresponds to higher-order syzygies among the 5 quadratic generators that are present in any MQ system with these dimensions. This confirms that the forgery system, after Weil descent, behaves identically to a generic MQ system at the degrees we can test.

State-of-the-art estimates via cryptographic_estimators. While the Hybrid-F4 model gives a useful analytical bound, the `cryptographic_estimators` library [3] provides state-of-the-art complexity estimates using optimized algorithms such as PXL (Parallel XL), BooleanSolveFXL, Crossbred, and HybridF5. Table 13 reports the fastest algorithm and its time complexity for each security level, alongside our analytical Hybrid-F4 estimate and the naive XL benchmark.

Interpretation. The `cryptographic_estimators` library gives the most efficient attack, significantly lower than our analytical Hybrid-F4 estimate (2^{154} vs. 2^{196} for Level I). This is because the library uses optimized algorithms (PXL for Levels I and III, BooleanSolveFXL for Level V) that improve upon the basic Hybrid-F4 model. However, **all estimates remain above the security thresholds**: the tightest margin is Level I with ~ 26 bits above 2^{128} .

Table 13: Complexity of generic-MQ attacks on the $e \times 5e$ forgery. The `cryptographic_estimators` library provides the most accurate estimate, using algorithms like PXL and BooleanSolveFXL. All estimates use $\omega = 3$.

Level	p	e	<code>cryptographic_estimators</code>		Hybrid-F4 (analytical)		Naive XL	Target
			Best algo	Time	W_{Hybrid}	$W_{e \times e}$	W_{XL}	
I	31	61	PXL	$\sim 2^{154}$	$\sim 2^{196}$	$\sim 2^{358}$	$\sim 2^{457}$	2^{128}
III	61	89	PXL	$\sim 2^{236}$	$\sim 2^{298}$	$\sim 2^{525}$	$\sim 2^{575}$	2^{192}
V	61	127	BoolSolveFXL	$\sim 2^{327}$	$\sim 2^{417}$	$\sim 2^{752}$	$\sim 2^{713}$	2^{256}

Remark 6. Among all forgery strategies, the `cryptographic_estimators` PXL algorithm on the $e \times 5e$ system gives the closest estimate to the security threshold: 2^{154} for Level I (margin ~ 26 bits), 2^{236} for Level III (margin ~ 44 bits), and 2^{327} for Level V (margin ~ 71 bits). This is the relevant state-of-the-art benchmark for assessing forgery resistance. The diagonal structure of SALAD does not introduce additional weaknesses: the experimental verification (Section 6.4) confirms that the forgery system behaves identically to a generic MQ system at degree $d = 4$.

Remark 7. The tight margin at Level I (~ 26 bits) warrants attention. The PXL algorithm is a parallel variant of XL that exploits modern matrix multiplication algorithms and parallelism. While 2^{154} remains computationally infeasible, the margin is smaller than typical cryptographic safety margins (~ 30 – 40 bits). This suggests that Level I parameters should be treated with caution, and that the hardened schedules (Section 3.2), which increase the linear span L by 50–67%, play an important role in ensuring that the Frobenius structure does not make the system easier than a generic MQ system.

Remark 8. The UOV-specific attacks (`CollisionAttack`, `KipnisShamir`) were also evaluated via the `cryptographic_estimators` UOV estimator. The `DirectAttack` (equivalent to MQ) is always the fastest forgery attack. `CollisionAttack` gives 2^{164} for Level I (vs. 2^{154} for `DirectAttack`), and `Kipnis-Shamir` key recovery gives 2^{934} (far above any threshold). The `WedgeAttack` does not apply (marked - by the estimator).

6.5 Key Recovery System: Detailed Results

The key recovery system consists of $5e$ quadratic equations in $4e$ variables over $\mathbb{F}_p$ (5 equations over $\mathbb{F}_q$, each expanded to e equations; $X_0 = 1$ leaves 4 unknowns). The equation-to-variable ratio is 5:4, so the system is slightly overdetermined. Table 14 gives the Macaulay dimensions.

Table 14: Macaulay matrix dimensions for the key recovery system ($m = 5e$ eqs, $n = 4e$ vars).

p	e	$d=2$ (rows $\times$ cols)	$d=3$ (rows $\times$ cols)	$d=4$ (rows $\times$ cols)
3 / 31	5	25×231	525×1771	5775×10626
3 / 31	7	35×435	1015×4495	15225×35960
3	11	55×1035	2475×16215	56925×194580
3	13	65×1431	3445×26235	93015×367290

At $d = 4$, the matrices (5775×10626 for $e = 5$, 15225×35960 for $e = 7$, 56925×194580 for $e = 11$) are too large for exact elimination. For $e \geq 11$ at $d = 3$, we employ a sparse Gaussian elimination

Table 15: Macaulay rank for the key recovery system (5e eqs in 4e vars, planted instances). Ranks for $e \geq 11$ at $d = 3$ computed via sparse Gaussian elimination. At $d = 4$, “def” decomposes as block-redundancy (diagonal-specific) + Koszul (generic). The dense $d=4$ rank for $e = 5$ is the predicted value (Koszul only, not directly computed).

p	e	Diagonal (SALAD)			Dense (UFOs)		
		$d=2$	$d=3$	$d=4$	$d=2$	$d=3$	$d=4$
3	5	20/25 (def=5)	420/525 (def=105)	4430/5775 (def=1345) = 1155+190	25/25 (def=0)	525/525 (def=0)	5475/5775* (def=300*) = 0+300
3	7	35/35 (def=0)	1015/1015 (def=0)	—	35/35 (def=0)	1015/1015 (def=0)	—
31	5	20/25 (def=5)	420/525 (def=105)	—	25/25 (def=0)	525/525 (def=0)	—
31	7	35/35 (def=0)	1015/1015 (def=0)	—	35/35 (def=0)	1015/1015 (def=0)	—
3	11	55/55 (def=0)	2475/2475 (def=0)	—	55/55 (def=0)	2475/2475 (def=0)	—
31	11	55/55 (def=0)	2475/2475 (def=0)	—	55/55 (def=0)	2475/2475 (def=0)	—
3	13	65/65 (def=0)	3445/3445 (def=0)	—	65/65 (def=0)	3445/3445 (def=0)	—
31	13	65/65 (def=0)	3445/3445 (def=0)	—	65/65 (def=0)	—	—

*Predicted from Koszul syzygies $\binom{r}{2}$ where r is the rank at $d=2$. At $e = 5$, diagonal $r = 20$: $\binom{20}{2} = 190$ Koszul +1155 block-redundancy = 1345. Dense $r = 25$: $\binom{25}{2} = 300$ Koszul, no block-redundancy.

algorithm (Section 6.7) that exploits the $< 2\%$ density of the Macaulay matrices to compute ranks efficiently.

6.6 Structure of the Diagonal-Specific Deficiencies at $e = 5$

The rank deficiencies observed for the diagonal key recovery system at $e = 5$ have two distinct origins: (1) *block-redundancy syzygies*, which are specific to the diagonal structure and constitute a potential weakness, and (2) *Koszul syzygies*, which are generic and appear in any quadratic system regardless of structure. We separate these two contributions carefully.

Degree-2 deficiency (block-redundancy, diagonal-specific). At $d = 2$, the Macaulay matrix has 25 rows and 231 columns. The rank is 20, giving a deficiency of $5 = e$. The dense system achieves full rank (25/25) at this degree, so this deficiency is entirely diagonal-specific. Since the 25 rows correspond to 5 blocks of $e = 5$ equations each, a deficiency of exactly e means that one entire block’s e equations are linearly dependent on the remaining $4e = 20$ equations. This is consistent with the following observation: in the diagonal case, each block k contributes the equation $\sum_{i=0}^4 c_{k,i} \Lambda_i^{p^{a_k} + p^{b_k}} = 0$. With $\Lambda_0 = 1$ fixed, the constant term $c_{k,0}$ is determined by the other coefficients and the planted Λ . The univariate structure means that the quadratic form of each block involves only 4 independent variable-groups (the coordinates of $\Lambda_1, \dots, \Lambda_4$), each of size

e . The oil condition $P_k(\Lambda) = 0$ imposes one $\mathbb{F}_q$ -linear relation per block, which descends to e $\mathbb{F}_p$ -linear relations. When the Frobenius structure is sufficiently degenerate (small e), one block's relations become linearly dependent on the others.

Degree-3 deficiency (block-redundancy, diagonal-specific). At $d = 3$, the Macaulay matrix has 525 rows and 1771 columns. The rank is 420, giving a deficiency of 105. The dense system achieves full rank (525/525), so this deficiency is again entirely diagonal-specific. It factorizes as $105 = 5 \times 21 = e \times \binom{4e+1}{1}$. The factor $\binom{4e+1}{1} = 21$ is the number of degree-1 multipliers per equation (monomials of degree ≤ 1 in $4e = 20$ variables: $1 + 20 = 21$). So the deficiency is exactly e times the number of multipliers, confirming that the same e -dimensional block-redundancy persists at degree 3: the dependent block's multiplied equations remain dependent on the others.

Degree-4 deficiency (block-redundancy + Koszul). At $d = 4$, the Macaulay matrix has 5775 rows and 10626 columns. The rank is 4430, giving a deficiency of 1345. Unlike degrees 2 and 3, this deficiency has *two* origins:

1. **Block-redundancy syzygies** ($1155 = 5 \times 231$): The 5 linear syzygies at degree 2 (one redundant block) persist through degree-2 multipliers, giving $5 \times \binom{4e+2}{2} = 1155$ dependent rows. This is the **diagonal-specific weakness**—the same mechanism as at $d \leq 3$.
2. **Koszul syzygies** ($190 = \binom{20}{2}$): These are **generic syzygies** of the form $g_j \cdot g_i - g_i \cdot g_j = 0$, present in *any* system of quadratic equations, regardless of structure. They arise from the 20 independent equations (after removing the 5 redundant ones) and contribute $\binom{20}{2} = 190$ additional null vectors. The dense system also has Koszul syzygies—in fact more of them ($\binom{25}{2} = 300$), since all 25 equations are independent.

We verified this decomposition explicitly: all $300 = \binom{25}{2}$ Koszul syzygy vectors (for all pairs of the 25 equations) were constructed and confirmed to lie in the null space of the Macaulay matrix. Their rank is 290 (not 300) because 10 syzygies involving the 5 dependent equations overlap with the block-redundancy space. The total null space dimension is $1155 + 290 - 100 = 1345$, where 100 is the overlap.

Implication for the dense case. The dense system has 25 independent equations (no block redundancy) and thus $\binom{25}{2} = 300$ Koszul syzygies at $d = 4$, giving a predicted deficiency of 300. The 190 Koszul syzygies in the diagonal case are therefore **not a diagonal-specific weakness**: they are generic and present in both variants. The **only** diagonal-specific excess is the block-redundancy component (1155), which is the same phenomenon observed at $d = 2$ and $d = 3$.

Remark 9. The Koszul syzygies $g_j g_i - g_i g_j = 0$ are a fundamental feature of the Macaulay matrix at degree $d \geq 4$ for any quadratic system. They are **not a weakness** of any particular structure: they affect diagonal and dense systems alike. The block-redundancy formula $e \times \binom{4e+d-2}{d-2}$ captures only the diagonal-specific component; at $d = 4$, generic Koszul syzygies add $\binom{r}{2}$ extra dimensions, where r is the number of independent equations. The fair comparison between diagonal and dense subtracts the generic component: $\text{def}_{\text{diag}} - \binom{r_{\text{diag}}}{2} = 1345 - 190 = 1155$ (diagonal-specific block redundancy) versus $\text{def}_{\text{dense}} - \binom{r_{\text{dense}}}{2} = 300 - 300 = 0$ (no block redundancy).

Ratio analysis. The deficiency ratio is $5/25 = 1/5$ at $d = 2$ and $105/525 = 1/5$ at $d = 3$, both equal to $1/\ell = 1/5$. This means exactly one out of five blocks is algebraically redundant in the

diagonal case at $e = 5$. The dense case has ratio $0/25 = 0$ at $d = 2$ and $0/525 = 0$ at $d = 3$, confirming no redundancy.

Disappearance at $e \geq 7$. At $e = 7$, the deficiency vanishes for both diagonal and dense. The Macaulay matrix has 35 rows and 435 columns at $d = 2$, and 1015 rows and 4495 columns at $d = 3$. Both variants achieve full rank. We further verified this convergence at $e = 11$ (55×1035 at $d = 2$, 2475×16215 at $d = 3$, for both $p = 3$ and $p = 31$) and $e = 13$ (65×1431 at $d = 2$ for both primes; 3445×26235 at $d = 3$ for $p = 3$ both variants and $p = 31$ diagonal). The dense case at $p = 31$, $e = 13$, $d = 3$ was not computed due to time constraints (matrix too dense for sparse elimination at $\sim 4\%$), but all other 32 tested configurations confirm full rank. The transition from $e = 5$ to $e \geq 7$ is sharp and persistent: the Frobenius structure becomes sufficiently generic at $e \geq 7$ to break all inter-block dependencies, and this genericity is maintained as e grows.

Multi-seed reproducibility. To confirm that the deficiencies are structural rather than coefficient-dependent, we ran 10 independent instances with different random seeds ($p = 3$, $e = 5$, diagonal, seeds 0–9). The results are perfectly reproducible:

- $d = 2$: $\text{def} = 5$ for all 10 seeds.
- $d = 3$: $\text{def} = 105$ for all 10 seeds.
- $d = 4$: $\text{def} = 1345$ for all 10 seeds.

Similarly, the disappearance at $e \geq 7$ is confirmed across multiple seeds: 10 seeds for $e = 7$ ($\text{def} = 0$ at $d = 2, 3$), 6 seeds for $e = 11$ ($\text{def} = 0$ at $d = 2, 3$), and 3 seeds for $e = 13$ ($\text{def} = 0$ at $d = 2, 3$). In total, 62 independent instances confirm the pattern.

Remark 10. The deficiency pattern $e \times \binom{4e+d-2}{d-2}$ (where $n = 4e$) is a diagnostic of the diagonal-specific block-redundancy. It is exact at $d \leq 3$ (where no generic syzygies exist) and a lower bound at $d = 4$ (where generic Koszul syzygies $\binom{r}{2}$ add to the total). Its presence at $e = 5$ and absence at $e \in \{7, 11, 13\}$ provides a concrete experimental criterion: if the block-redundancy is nonzero at the target e , the diagonal structure introduces a weakness; if zero, it does not. The convergence at $e = 7$ is confirmed at $e = 11$ and $e = 13$ across 62 tested instances, strongly suggesting that the block-redundancy is zero for all $e \geq 7$, including the real UFOs parameters $e = 61, 89, 127$.

6.7 Sparsity and Performance

The Macaulay matrices are extremely sparse. Table 16 reports the number of nonzeros (nnz) and the density (nnz / (rows $\times$ cols)) for representative configurations.

Table 16: Sparsity of Macaulay matrices for the forgery system ($e = 5$, $p = 31$).

Variant	$d=2$	$d=3$	$d=4$
Diagonal	340 nnz (19.4%)	8840 nnz (2.1%)	119340 nnz (0.3%)
Dense	1511 nnz (86.1%)	39286 nnz (9.2%)	530361 nnz (1.3%)

The diagonal variant is approximately 4–5 $\times$ sparser than the dense variant, reflecting the absence of cross-terms. At degree $d = 2$, the diagonal matrix has 340 nonzero density (19.4%) versus 1511

(86.1%) for dense. The density drops rapidly with degree: at $d = 4$, both variants are below 1.5% dense, but the diagonal variant remains $\sim 4.5\times$ sparser. Despite this sparsity difference, the rank profiles are identical for the forgery system (Table 12), confirming that sparsity alone does not create algebraic weaknesses.

For the key recovery system at $e = 7$, $d = 3$ (1015×4495), the diagonal variant has 76792 nonzeros density (1.7%) and requires ~ 3 s for exact Gaussian elimination, while the dense variant has 297540 nonzeros density (6.5%) and requires ~ 21 s—a $7\times$ speedup from sparsity.

For $e = 11$, $d = 3$ (2475×16215), sparse Gaussian elimination takes ~ 9 s (diagonal, 437175 nnz, 1.1%) versus ~ 97 s (dense, 1679355 nnz, 4.2%)—an $11\times$ speedup. For $e = 13$, $d = 3$ (3445×26235), the diagonal case takes ~ 49 s (836552 nnz, 0.9%) and the dense case ~ 271 s (3296653 nnz, 3.7%). The sparsity advantage of the diagonal variant grows with e , making sparse-aware algorithms increasingly effective. The rank computation time scales as $O(\text{nnz} \cdot \text{rank})$, making sparse-aware algorithms essential for larger e .

6.8 Interpretation

Forgery. The forgery system (Table 12) shows **no degree drops** for diagonal or dense, at any parameter. The Macaulay matrix has full row rank at $d = 2$ and $d = 3$ in all cases. The system is extremely underdetermined (e equations in $5e$ variables), so full row rank is the generic expectation. The diagonal structure does not introduce any additional dependencies. This confirms that diagonal restriction does not weaken forgery resistance.

Key recovery. The key recovery system (Table 15) reveals a more nuanced picture. At $e = 5$, the diagonal system exhibits rank deficiencies in all degrees tested. At $d = 2$ and $d = 3$, these deficiencies (5 and 105 respectively) are **entirely diagonal-specific**: the dense system achieves full rank. They arise from the univariate structure, which creates linear relations among the $5e$ equations—one block out of five becomes algebraically redundant. At $d = 4$, the deficiency (1345) is a mixture: 1155 is the same diagonal-specific block-redundancy, but 190 is the generic Koszul syzygies that the dense system also exhibits (with $300 = \binom{25}{2}$ of its own). The **diagonal-specific weakness** at $d = 4$ is therefore 1155, not 1345.

At $e = 7$, the block-redundancy disappears: both diagonal and dense achieve full rank at $d = 2$ and $d = 3$. (At $d = 4$, both would still have Koszul syzygies, but these are generic and not a weakness of either structure.) The transition from $e = 5$ to $e \geq 7$ is sharp: the Frobenius structure becomes sufficiently generic to break the inter-block dependencies.

Remark 11. *For the real UFOs parameters ($e = 61, 89, 127$), the diagonal structure is expected to behave identically to the dense case. The block-redundancy at $e = 5$ is a small-parameter artifact that disappears at $e \geq 7$; it is not a structural weakness. The Koszul syzygies in $d \geq 4$ are generic and affect both variants equally.*

Conclusion on the diagonal structure. We distinguish two types of rank deficiencies:

1. **Generic syzygies (Koszul):** present at $d \geq 4$ in *any* quadratic system, both diagonal and dense. They are not a weakness of any particular structure.

2. **Diagonal-specific block** redundancy: present in $e = 5$ for all $d \geq 2$, absent in $e \geq 7$. This is the only potential weakness of the diagonal structure and disappears at the security-level parameters.

The fair comparison between SALAD and UFOs removes the generic component. At $e = 5$, $d = 4$: the diagonal-specific excess is $1345 - 190 = 1155$ (block redundancy), versus $300 - 300 = 0$ for dense. At $e \geq 7$, both variants have zero diagonal-specific excess at all degrees tested.

We find **no evidence** that the diagonal structure of SALAD introduces weaknesses beyond those of the generic UFOs scheme.

1. For forgery, diagonal and dense systems have identical rank profiles at all tested parameters.
2. For key recovery, diagonal-specific block-redundancy in $e = 5$ (def = 5 in $d = 2$, 105 in $d = 3$, 1155 in $d = 4$ after subtracting generic Koszul syzygies) is absent in the dense system, but disappears in $e = 7$ and is expected to be absent at security-level parameters.
3. Generic Koszul syzygies in $d \geq 4$ affect both variants equally and are not a structural weakness.
4. The attack paths (univariate span, bivariate resultant, descent of Weil + XL) are identical for both variants, as established in Section 3.1.

7 UFOs Parameters and Performance

The UFOs scheme (and its SALAD diagonal variant) uses the same parameter structure: $o = m = 1$, $v = 4$, $n = 5$, $\ell = 5$ Frobenius blocks, with field parameters (p, e) chosen per NIST security level. Tables 17 and 18 report the concrete key sizes and reference timings from the UFOs specification [1], measured on the Online Magma Calculator.

Table 17: Concrete parameters and key/signature sizes (bytes) for UFOs/SALAD [1].

Level	p	e	$ \sigma $	Pub	$ \sigma + \text{Pub} $	Sec	SeedPub	$ \tau $
I	31	61	223	207	430	32	16	32
III	61	89	366	350	716	32	16	32
V	61	127	509	493	1002	32	16	32

The secret key consists of a 32-byte seed (**SeedSec**), from which all secret material is derived. The public key combines a 16-byte seed (**SeedPub**) with the published $o \times o$ blocks A_{rk}^0 , totaling $mo^2e\ell$ field elements. The signature $\sigma = (\Sigma, \tau)$ contains $n \cdot e$ field elements plus a 256-bit salt.

Table 18: Reference timings from the Online Magma Calculator [1]. $o=1, v=4, \ell=5$.

Level	p	e	KeyGen	Sign	Verify
I	31	61	1.7 ms	7.1 ms	1.0 ms
III	61	89	2.9 ms	13.1 ms	1.7 ms
V	61	127	3.2 ms	20 ms	2.1 ms

Signing is dominated by a single $e \times e$ linear solve over $\mathbb{F}_p$ (after fixing the vinegar variables), while verification requires evaluating $\ell = 5$ Frobenius-block products over $\mathbb{F}_q$. Key generation

requires expanding the seeds and computing the missing oil–oil blocks via the isotropy constraint. All operations are very fast (sub-25 ms at all levels), placing UFOs/SALAD in the same performance niche as the most compact post-quantum MQ signatures.

8 Implementation Notes

- **The exact rank over $\mathbb{F}_p$:** the SVD-based rank and Gram matrix approaches ($G = C^T C$) give incorrect results for finite-field matrices due to isotropic vectors. Use exact Gaussian elimination mod p .
- **Deterministic seeds:** Python’s `hash()` is non-deterministic across sessions (hash randomization since Python 3.3). Always use deterministic seeds (e.g. `random.Random(seed · 1000 + constant)`) for reproducible experiments.
- **Integer overflow:** For $e \geq 17$, `rng.integers(0, p**e - 1)` overflows `int64`. Use `random.randrange(0, p**e - 1)`.
- **Performance:** Custom NumPy Gaussian elimination handles 2000×400 in ~ 3.5 s, 5000×800 in ~ 35 s. The constraint matrix C in $e = 61$ (39711×3721) is too large for precise dense elimination in Python; use the U_{dim} criterion as a proxy for degree-0 syzygies.
- **galois library:** The `galois` library (v0.4.11) provides `GF(p**e)` with `primitive_element` and `.vector()` API. The `.vector()` method returns coordinates in reverse order $[\alpha^{e-1}, \dots, \alpha^0]$; reverse with `vec[::-1]`. SageMath is not available.
- **Frobenius vector precomputation:** For large e , precompute $\phi_a(\alpha^j) = \alpha^{j \cdot p^a}$ for all distinct schedule exponents once, then reuse across triplets. Cost: ~ 1 s for $e = 61$.

9 Conclusion

The diagonal variant SALAD is secure at all NIST security levels. The diagonal structure eliminates cross-terms, making the monomials univariate and the system separable. However, Frobenius powers give the univariate polynomial a degree of p^{54} over $\mathbb{F}_q$, making direct factoring infeasible. After descent of Weil, the quadratic system has a degree of solution $\gtrsim 55$, which yields work $\gtrsim 2^{337}$ for Level I – well above the security bound 2^{128} .

The XL solving degree analysis reveals that the forged 3-term Frobenius-sum equation, after descent of Weil, behaves as a generic MQ system for generic coefficients. The constraint matrix C has full column rank for $\sim 80\%$ of the coefficient choices in small e , and the well-spread UFOs schedule ensures $U_{\text{dim}} = e$ (no degree-0 syzygies) in all parameters tested, including $e = 61$. The phenomenon $D_{\text{lin}} = 3$ is coefficient-dependent, not structural, and is expected to vanish at large e . The degree of XL solution for generic coefficients is $\gtrsim e$, which gives work $\gtrsim 2^{362}$ at Level I.

The comparison between diagonal (SALAD) and dense (UFOs) structures provides the following detailed findings:

- **Forgery system** (e eqs in $5e$ vars): At all parameters tested ($p \in \{3, 31\}$, $e \in \{5, 7, 11\}$), the Macaulay matrix achieves full row rank at degrees $d = 2$ and $d = 3$ for both variants. For $e = 5$,

the matrix $d = 3$ is 130×3276 with rank 130/130 (def = 0). For $e = 11$, the matrix $d = 3$ is 616×30856 with rank 616/616 (def = 0). No degree falls are observed. The diagonal structure introduces zero additional syzygies. At $d = 4$ ($e = 5$), all three variants (diagonal, dense, generic) exhibit the identical deficiency of 140, confirming that the forgery system behaves as a generic MQ system (Section 6.4). The state-of-the-art `cryptographic_estimators` library (Section 6.4) gives the closest estimate to the security threshold: 2^{154} for Level I (PXL algorithm, margin ~ 26 bits), 2^{236} for Level III (PXL, margin ~ 44 bits), and 2^{327} for Level V (BooleanSolveFXL, margin ~ 71 bits). All estimates remain above the security bound.

- **Key recovery system** ($5e$ eqs in $4e$ vars): At $e = 5$, the diagonal system exhibits rank deficiencies at all degrees tested. At $d \leq 3$, these are **entirely diagonal-specific** (block-redundancy): def = $5 = e$ at $d = 2$ (rank 20/25), def = $105 = e \times \binom{4e+1}{1}$ at $d = 3$ (rank 420/525), absent in dense. At $d = 4$, the deficiency (def = 1345, rank 4430/5775) decomposes into 1155 block-redundancy (diagonal-specific) + 190 Koszul syzygies $\binom{20}{2}$ (generic, also present in dense with $\binom{25}{2} = 300$). The **diagonally specific excess** at $d = 4$ is 1155, which matches the block-redundancy pattern. At $e = 7$, block-redundancy vanishes: both variants achieve full rank at all degrees tested (35/35 at $d = 2$, 1015/1015 at $d = 3$). Confirmed at $e = 11$ (55/55 at $d = 2$, 2475/2475 at $d = 3$, for both $p = 3$ and $p = 31$) and $e = 13$ (65/65 at $d = 2$ for both primes; 3445/3445 at $d = 3$ for $p = 3$ both variants and diagonal $p = 31$). All deficiencies are perfectly reproducible across 10 random seeds (62 total instances), confirming their structural nature.
- **Sparsity**: diagonal Macaulay matrices are $4\text{--}5\times$ sparser than dense ones. For the forgery system at $e = 5$, $p = 31$, the diagonal $d = 2$ matrix has 340 nonzeros (19.4%) versus 1511 (86.1%) for dense. At $d = 3$, the diagonal matrix has 8840 nnz (2.1%) versus 39286 (9.2%). For the key recovery system, sparse Gaussian elimination yields a $7\times$ speedup at $e = 7$ (~ 3 s vs ~ 21 s), $11\times$ at $e = 11$ (~ 9 s vs ~ 97 s) and $5.5\times$ at $e = 13$ (~ 49 s vs ~ 271 s). The sparsity advantage of the diagonal variant increases with e .

No evidence is found that the diagonal structure of SALAD introduces weaknesses beyond those of the generic UFOs scheme. The block-redundancy at $e = 5$ is a small-parameter artifact, disappearing at $e \geq 7$ and confirmed absent at $e = 11$ and $e = 13$ in all instances tested 62. Generic Koszul syzygies in $d \geq 4$ affect both variants equally and are not a structural weakness. The convergence is robust across primes ($p = 3$ and $p = 31$), degrees ($d = 2, 3, 4$), and random seeds, strongly suggesting that the pattern is maintained in the parameters of the security-level ($e = 61, 89, 127$).

We further propose improved hardened schedules (Section 3.2) that increase the linear span L by 50–67% compared to the original Bariant schedules, and introduce new window diagnostics w_t and Bézout bounds B_t that provide additional security margins. The Bézout bound B_4 yields work $\gtrsim 2^{337}$ for Level I, consistent with the XL solving degree estimate. These schedules apply to both UFOs and SALAD.

A critical methodological lesson is that non-deterministic Python `hash()` seeds can produce misleading experimental results. The initial universal finding $D_{\text{lin}} = 3$ was an artifact of this issue, corrected by using deterministic seeds.

Future work should:

1. Develop efficient algorithms to compute the rank of the constraint matrix in $e = 61$ (e.g., through modular sparse linear algebra or probabilistic methods).

2. Characterize the algebraic conditions under which “special” coefficients trigger rank deficiencies, and quantify the probability as a function of e and p .
3. Investigate whether the block-diagonal structure after descent of Weil introduces additional algebraic shortcuts beyond the XL framework.
4. Study whether other structured restrictions (sparse, banded, low-rank public matrices) preserve security.
5. Verify the disappearance of diagonal-specific key recovery deficiencies at intermediate values ($e = 9, 11, 13$) to strengthen the convergence claim. **Done:** $e = 11$ and $e = 13$ tested, both confirm full rank for diagonal and dense.

References

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